

Gamma detection for source localisation in targeted alpha therapy

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Abstract. Targeted alpha therapies can deliver a highly localised dose, but treatment planning and delivery depend on whether the radionuclide source remains near the intended site after administration. In the AlphaGlue delivery concept, alpha emitters are applied to a tumour region in a thin glue layer. We present a Geant4 simulation study showing that decay-chain gamma detection can distinguish retained, broadly diffused and locally displaced source states in an AlphaGlue-like geometry. Using 50 independent repetitions, baseline-subtracted gamma-hit profiles reconstructed the broad diffusion fraction with $R^2 = 0.9987$ and a mean absolute error of about one percentage point. Localised-source residuals identified displaced activity with mean position errors of 16–30 mm. These results support gamma detection as a candidate method for emitter-position monitoring, treatment planning and delivery verification.

1 Introduction

In targeted alpha therapy, the delivered dose is strongly local. A treatment plan therefore depends not only on the initial activity distribution, but also on whether source material and progeny remain near the intended deposition site, diffuse into surrounding tissue or are re-distributed by biological transport [2, 3]. The AlphaGlue concept proposes suspending alpha emitters in a thin glue layer applied to tissue [1]. The concept has previously been evaluated with Geant4 and Geant4-DNA simulations for dose and DNA-damage modelling [4, 6]. This study investigates whether the accompanying gamma signal can be used as a theranostic observable to monitor emitter position, detect source movement and provide useful information as a function of detection time.

2 Simulation model

Photon transport was simulated with the Geant4 toolkit [4]. The model places a 40 mm × 2 mm × 40 mm loaded glue layer on the +y surface of a 20 cm × 20 cm × 150 cm water phantom. Six detector panels surround the phantom at 20 cm from the water surface. Each photon entry into a detector panel is counted as one hit.

Seven source configurations were studied to separate three clinically relevant possibilities: emitter retained in the loaded glue layer, emitter broadly mixed into the surrounding volume and emitter transported to a localised displaced region. The retained-source case is used as the baseline. Some alpha emitters in the decay chain can be taken up by the blood, with large variation reported between experiments [2, 3]. This further emphasises the need for local dose verification. The simulations therefore include cases in which 10%, 50% or 90% of the radioisotope is distributed throughout the water volume, and cases in which 20%, 40% or 60% of the source is placed in a small displaced water region.

Each configuration was simulated with 50 independent repetitions of 10,000 primary decays, giving 500,000 simulated decays per configuration and 3.5 million simulated decays across all source states. The repeated runs define the run-to-run uncertainty used in the residual and detectability analysis. Where activity-scaled observables are quoted, the simulated decays are rescaled to a DaRT-like reference activity of 3 μCi (111 kBq), matching the order of a single DaRT seed [5].

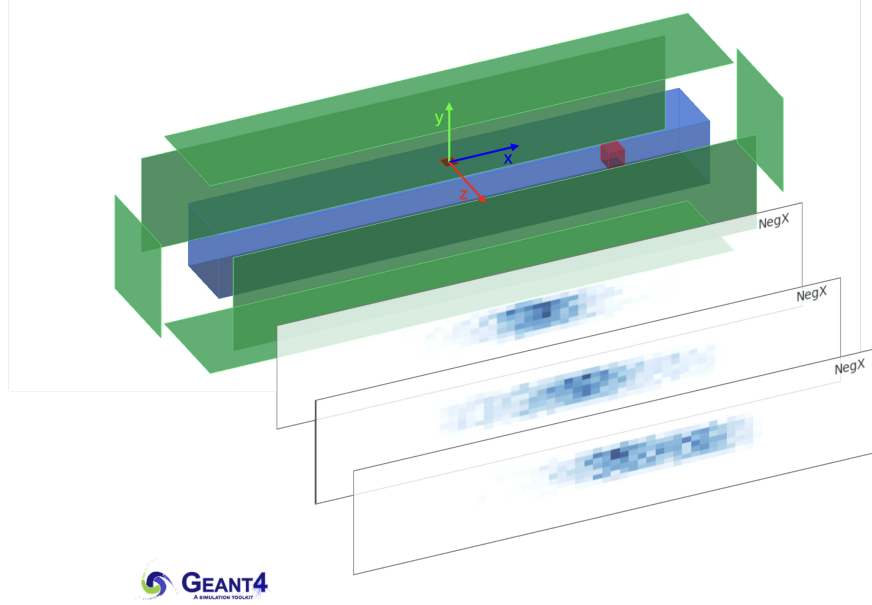


Figure 1: Simulation geometry showing the water phantom (blue), surrounding detector panels (green), a displaced source (red) and example detector projections.

3 Baseline subtraction

The total detector-entry rate changed only weakly between retained and redistributed source states. It changed by about 3.8% between the retained baseline and the 90% broad-diffusion case. In contrast, baseline-subtracted profiles show a clear spatial signature. The retained source defines the reference profile $\bar{h}_p^B(z)$ for detector panel p . Candidate source states are represented by

$$\Delta h_p(z) = h_p(z) - \bar{h}_p^B(z), \quad (1)$$

with the repeated baseline simulations defining the bin-wise noise scale. Positive residuals mark detector regions where the candidate produces more gamma entries than the retained-source case, while negative residuals mark depleted regions. Broad diffusion widens the residual pattern and changes the balance between the glue-facing and opposite detector panels. Localised displacement produces a shifted excess near the displaced source region.

Figure 2 shows that source movement is better detected from baseline-subtracted spatial hit patterns than from the total hit rate alone.

4 Residual-based source metrics

The residual analysis was organised into three source-monitoring metrics. First, diffusion was inferred from profile-shape features and panel-balance changes. The $+y$ panel response decreases as activity moves away from the glue layer, while the profile width and side-panel share

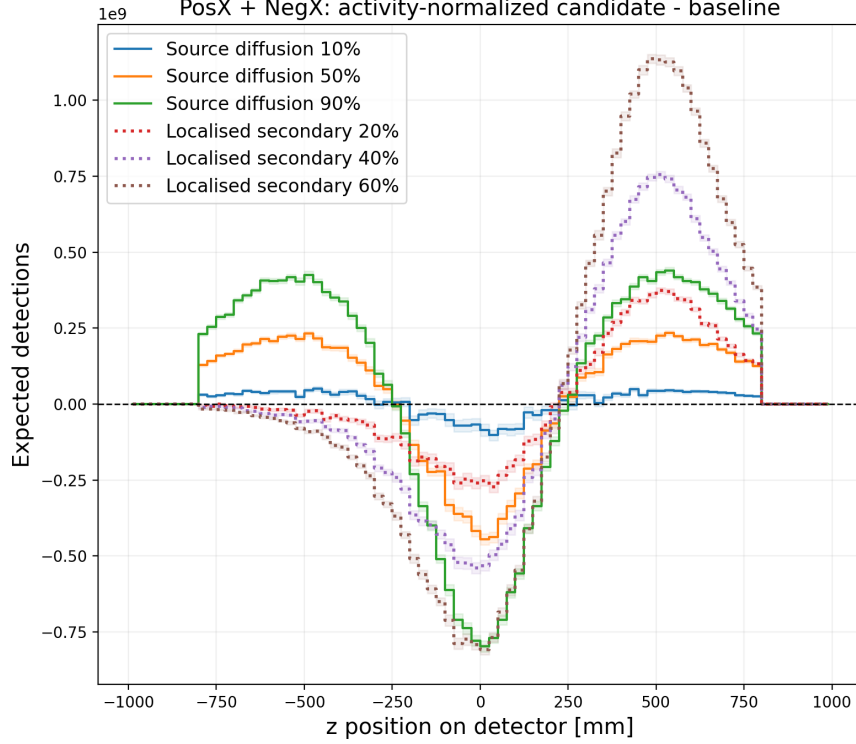


Figure 2: Activity-normalised, baseline-subtracted $+x$ and $-x$ detector profiles for broad diffusion and localised displacement. Shaded bands show variation across repetitions.

increase. Second, activity-retention observables used the central $+y$ region of interest (ROI) as an applicator-facing signal. This estimates activity still consistent with the retained-source geometry, but is interpreted only after a secondary-source test because compact displaced activity can also suppress the central signal. Third, compact secondary sources were searched for using positive residuals in a displaced-region ROI and panel-asymmetry changes that are small for broad diffusion but large for off-applicator localisation.

For the present geometry, the displaced-source search ROI is centred on the positive- z region containing the simulated compact water source. The secondary-source score integrates the positive residual in this ROI and compares it with the retained-source baseline variation. If this test is positive, the location is estimated from the positive-residual centroid on the detector views that best project the source onto the longitudinal axis. Diffusion fraction is reported for broad redistribution candidates, while localised cases are first reported as compact secondary-source detections with an estimated position.

5 Signal inference

Using profile-shape features from repeated simulations, the broad diffusion fraction was recovered with $R^2 = 0.9987$ and a mean absolute error of about one percentage point. A separate secondary-source metric identified the localised displaced source in all 20%, 40% and 60% localised-displacement cases. The reconstructed displaced-source positions were approximately $z = 516$ mm, 528 mm and 530 mm, respectively, compared with the simulated $z = 500$ mm source region. Across these localised fractions, mean position errors were 16–30 mm.

Profiles used for central signal estimate: ROI -150 to 150 mm

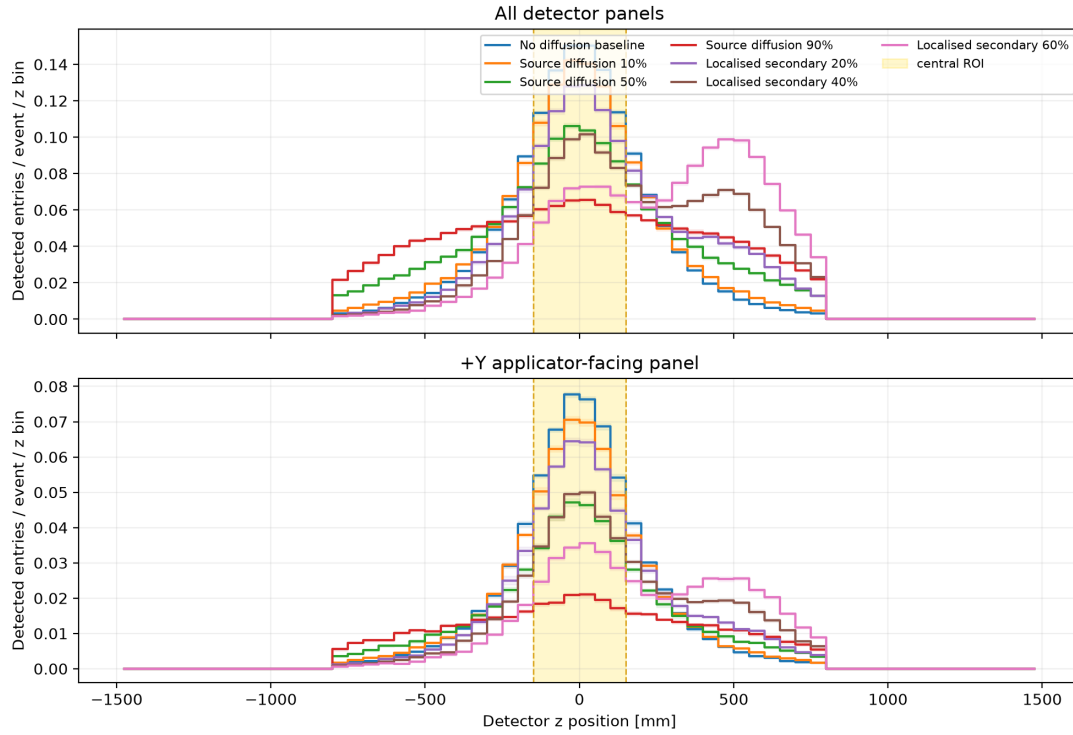


Figure 3: Central-ROI signal used to separate retained applicator response from activity redistributed outside the applicator region.

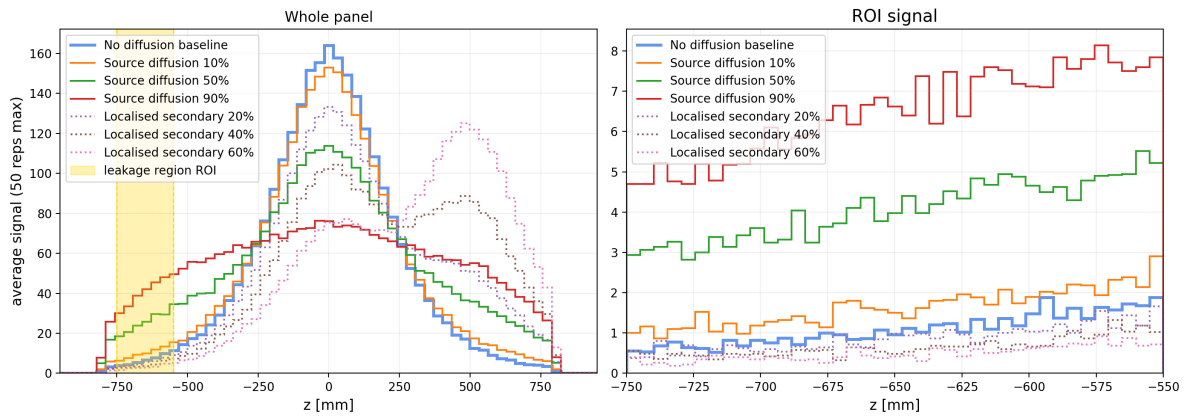


Figure 4: Whole x -panel profiles (left) and the displaced-region ROI signal (right). The ROI highlights the broad-diffusion excess outside the retained-source region.

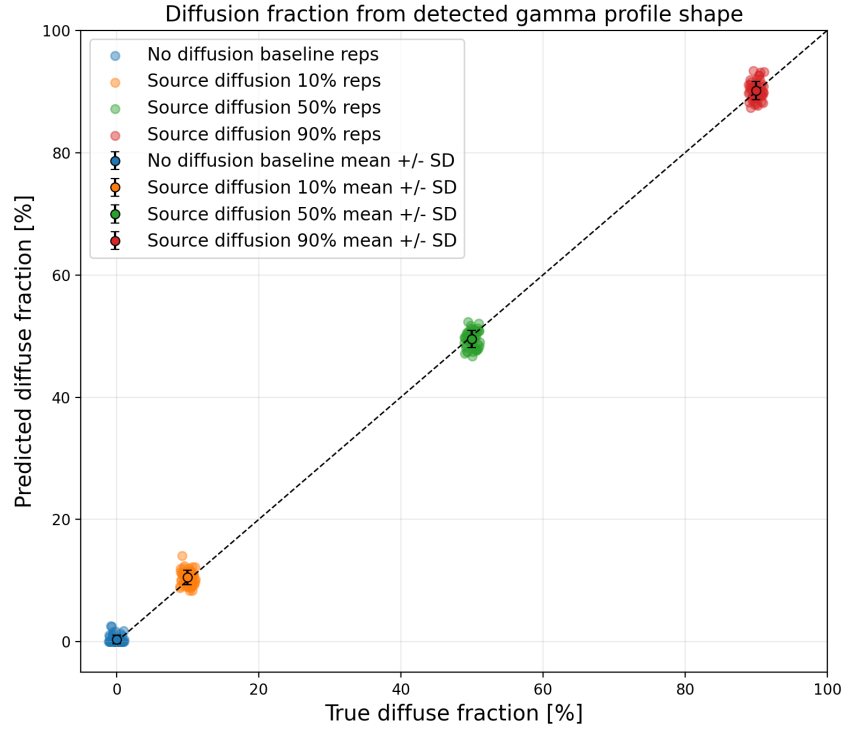


Figure 5: Diffusion fraction inferred from 50 independent repetitions. Small points show individual repetitions and large markers show the mean and standard deviation.

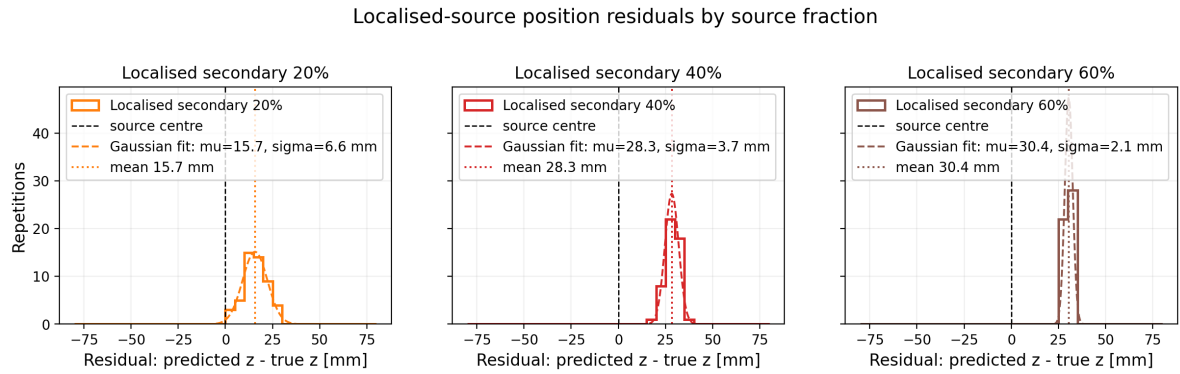


Figure 6: Localisation residuals across 50 repetitions for each displaced-source fraction. The dashed black line marks the true source centre; coloured lines indicate the mean and Gaussian fit.

6 Time dependence

Cumulative time-window profiles show that the baseline-subtracted signal is already visible on hour-scale windows and becomes stronger over longer observation windows. Across the 50 repetitions, detectability required candidate-baseline separation greater than 3σ of the run-to-run baseline variation. The 50–90% broad-diffusion and 20–60% localised-displacement cases crossed threshold in about 1–2 hours, while the weaker 10% broad-diffusion case required about 6 hours. A $t < 10^6$ s analysis cut suppresses a late decay-chain tail while retaining the early signal relevant to the simulated source-localisation window.

Table 1: First 3σ detection times from the time-window analysis of 50 independent repetitions.

Case group	First 3σ detection
10% broad diffusion	about 6 h
50–90% broad diffusion	about 1–2 h
20–60% localised displacement	about 1–2 h

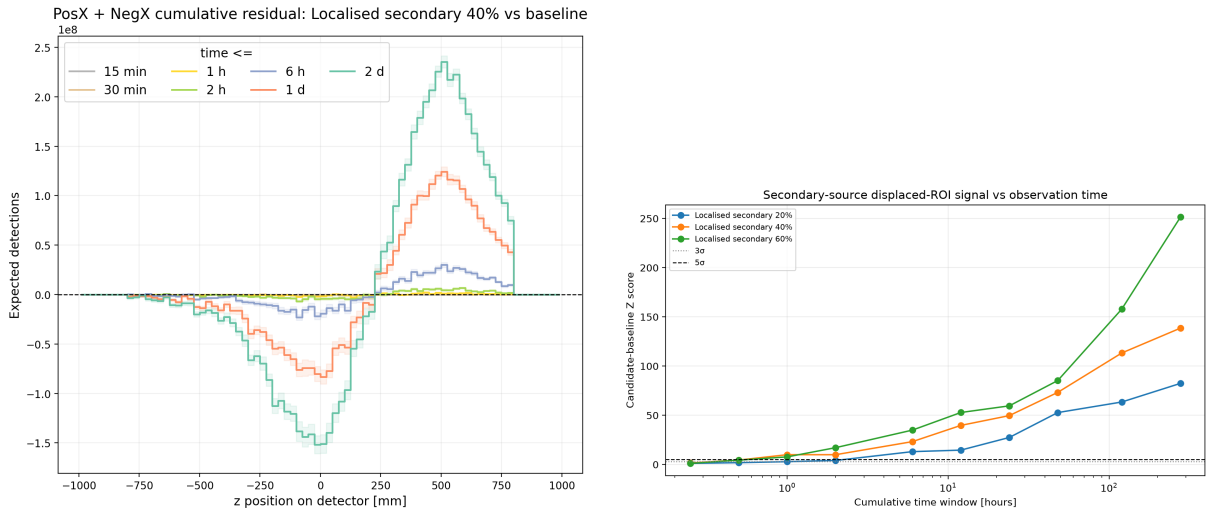


Figure 7: Time-dependent residual analysis. Left: cumulative $+x$ and $-x$ profile for 40% localised displacement, showing a growing excess near $z = 500$ mm. Right: significance of the distinct displaced-source ROI signal with observation time.

7 Conclusion

Decay-chain gamma detection provides source-location information that is not captured by total count rate alone. Baseline-subtracted profiles distinguish retained, broadly diffused and locally displaced activity, while residual features support quantitative estimates of diffusion fraction and displaced source position. Further work will include detector response, backgrounds and patient-scale geometries so that activity-normalised profiles can be interpreted as experimentally measurable count rates.

References

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